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Forgetting Through Transforming: Enabling Federated Unlearning via Class-Aware Representation Transformation

Qi Guo[1,†], Zhen Tian[1,†], Minghao Yao[1], Saiyu Qi[1,*], Yong Qi[1], Bingyi Liu[2]
[1]Xi'an Jiaotong University
[2]Wuhan University of Technology
qiguoqg@outlook.com, {zhentian, minghao_yao}@stu.xjtu.edu.cn
{saiyu-qi, qiy}@xjtu.edu.cn, byliu@whut.edu.cn

Abstract

Federated Unlearning (FU) should satisfy three key requirements: a guarantee of data erasure, preservation of model utility, and reduction of unlearning time. Recent studies focus on identifying and modifying original model parameters relevant to unlearning data. While they can achieve faster unlearning, they degrade the model performance on remaining data or fail to forget unlearning data due to the difficulty in isolating specific parameters of the unlearning data. By revisiting the representation distribution of the optimal unlearning models (i.e., the retrained models), we observe that unlearning data tends to cluster within semantically related categories of remaining data. This inspired us to transform the distribution of unlearning data to fuse with similar categories in the remaining data for effective FU. Based on this insight, we propose a novel framework, named FUCRT, to achieve Federated Unlearning via Class-aware Representation Transformation. FUCRT consists of two key components: (1) a transformation class identification strategy (TCI) that leverages the original model to identify appropriate transformation classes for unlearning data, and (2) a targeted transformation learning process (TTL) with cross-class fusion mechanism to ensure effective and consistent transformation. Extensive experiments on four datasets demonstrate that FUCRT not only achieves 100% of data erasure but also outperforms state-of-the-art methods by an average of 2.96% and 3.78% in utility preservation under IID and Non-IID settings, respectively. Moreover, it reduces unlearning time by 19.13%∼ 96.38%. Codes are at https://github.com/zhentian777/FUCRT.

1. Introduction

Federated Learning (FL) [33, 48] is a distributed learning paradigm that allows multiple devices to collaboratively train a global model without directly exchanging local data, thereby reducing the risk of data breaches. However, it remains necessary to remove specific classes of data from global models to comply with emerging data protection regulations [13]. Moreover, privacy legislation, including the European Union's *General Data Protection Regulation* (GDPR) [43] and the *Right to be Forgotten* [11], also grants users the right to remove specific classes of data from trained global models. Therefore, Federated Unlearning (FU) [45, 47], which can address this requirement, has gained increasing attention.

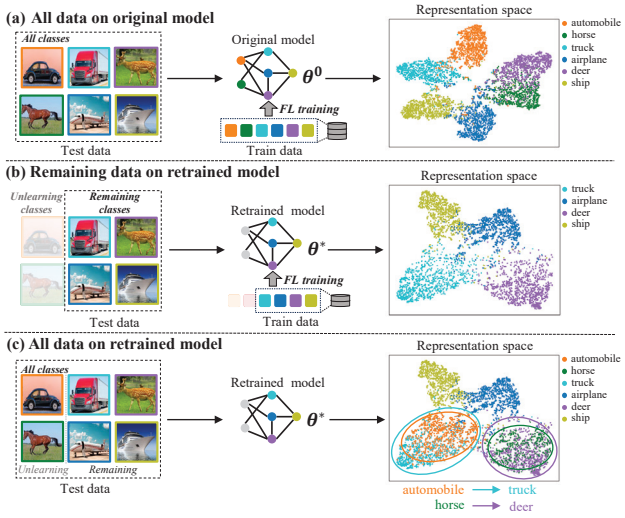

Figure 1. The distribution patterns of remaining and unlearning data in original and retrained models.

In FU, the original dataset can be divided into two parts, i.e., remaining data and unlearning data. Based on this, FU should satisfy three key requirements: ***i) erasing guarantee***, which ensures the effective forgetting of unlearning data; ***ii) model utility preservation***, which requires maintaining the performance of the remaining data on the unlearning model;

*corresponding author. † equal contribution.

iii) reducing unlearning time, which is essential for improving the efficiency of FL, specifically in scenarios where unlearning operations are requested frequently. A native approach for FU is to follow the standard FL training process and retrain the model from scratch with only the remaining data [29]. Although this way can achieve both the erasing guarantee and model utility preservation, it cannot be practically applied due to the high computational cost of retraining. To alleviate this overhead, recent studies focus on identifying and modifying original model parameters relevant to unlearning data, employing methods such as channel pruning [44], gradient ascent [45], and through auxiliary models [14, 51]. While they can achieve faster unlearning, the complex parameter interactions [2, 12] in deep neural networks make it difficult to identify and isolate parameters related to unlearning data from those associated with remaining data. As a result, these methods will inevitably degrade the performance of the remaining data or fail to forget unlearning data, i.e., impacting the utility preservation or undermining erasing guarantee. Thus, it is expected to realize FU ensures time efficiency, satisfies the erasing guarantee, and preserves model utility simultaneously.

To explore the underlying nature of FU, as shown in Fig. 1, we compare the distribution patterns of remaining and unlearning data in the representation space of two models: the original global model and the retrained global model trained from scratch with only remaining data. The performance of utility preservation is reflected by strong intra-class compactness and inter-class separability of the remaining data in representation space (see Fig. 1(b)), and the erasing effect is manifested in mixing unlearning data representations with the remaining data distribution (see Fig. 1(a) and (c)). Interestingly, we observe that in the representation space of the retrained model, unlearning data does not distribute randomly but tends to cluster within specific data related to their original categories. We further conduct empirical analysis, which reveals that just 2 semantically closest classes are sufficient to cover approximately 80% of unlearning instances on CIFAR10 dataset. For example, 76.24% and 9.2% of the data in *automobile* are forgotten into *truck* and *ship*, respectively. Therefore, unlike recent studies [9, 14] that presuppose a random distribution of unlearning data, we consider transforming the unlearning data into approximate or related categories within the remaining data for effective FU.

In this work, we propose a novel framework, named **FUCRT**, to achieve **F**ederated **U**nlearning via **C**lass-aware **R**epresentation **T**ransformation. **The core idea of FUCRT is the transformation, which is mainly reflected in two aspects:** (1) Considering that using a retrained model to determine appropriate target transformation classes for unlearning data is impractical in real-world FU scenarios, we design a **transformation class identification strategy (TCI)** that leverages the original model to identify appropriate target transformation classes for unlearning data. (2) We develop a **targeted transformation learning process (TTL)** for unlearning. To improve distribution integration between unlearning data and target transformation class data in the representation space, we devise a cross-class fusion mechanism during fine-tuning. This is because even when unlearning data appears completely erased to classifiers, it may still be detected by Membership Inference Attacks (MIA) if not tightly clustered with transformation class data in the representation space [10, 14, 39]. Additionally, this mechanism also incorporates auxiliary cross-client fusion to ensure consistent unlearning performance across all participating clients due to data heterogeneity [22] in FL. Unlike previous approaches that attempt to locate and modify model parameters associated with unlearning data, our FUCRT allows the model to adaptively adjust its parameters, achieving efficient erasure while better preserving model utility on the remaining data.

Our main contributions can be summarized as follows:

(1) We propose a novel idea for FU, i.e., transforming the distribution of unlearning data to fuse with their semantically related remaining data in representation space.
(2) Based on this idea, we propose a new framework, the FUCRT, with transformation class identification (TCI) and targeted transformation learning (TTL) to realize effective and efficient FU.
(3) Extensive experiments on four datasets demonstrate that FUCRT not only achieves effective data erasure but also outperforms state-of-the-art methods by an average of 2.96%/3.78% in utility preservation under IID/Non-IID settings, while reducing unlearning time by 19.13%∼96.38%.

2. Problem Definition

A federated system comprising K clients and one server, which learns a global model θ on dataset $\mathcal{D} = \{(x_i, y_i)|y_i \in \mathcal{C}\}_{i=1}^{|D|}$ collaboratively, where x_i, y_i, $\mathcal{C}$ and N denote an individual sample, its corresponding label, the total class set, and the number of all samples, respectively. For each client $k \in \{1, \ldots, K\}$, $\mathcal{D}_k = \{(x_i, y_i)|y_i \in \mathcal{C}_k\}_{i=1}^{|\mathcal{D}_k|}$ represents its local dataset, where $\mathcal{C}_k \subseteq \mathcal{C}$ and $|\mathcal{D}_k|$ denote the local class set and the number of local samples, respectively. Here, we have $|\mathcal{D}_1| + |\mathcal{D}_2| + \cdots + |\mathcal{D}_K| = |D|$ and $\mathcal{C}_1 \cup \mathcal{C}_2 \cup \cdots \cup \mathcal{C}_K = \mathcal{C}$. In this work, we focus on the class-level unlearning task. Let $\mathcal{C}_\mathcal{F} \subseteq \mathcal{C}$ denote the class set to be forgotten. Our goal is to adjust the global model θ into θ^R such that θ^R excludes the unlearning data $\mathcal{S} = \{(x_i, y_i)|y_i \in \mathcal{C}_\mathcal{F}\} \subseteq \mathcal{D}$ from its prediction space while preserving its utility on the remaining data $\mathcal{D}\backslash\mathcal{S}$. Therefore, for each client k, the local model aims to unlearn its subset $\mathcal{S}_k = \{(x_i, y_i)|y_i \in \mathcal{C}_k \cap \mathcal{C}_\mathcal{F}\} \subseteq \mathcal{D}_k$, which

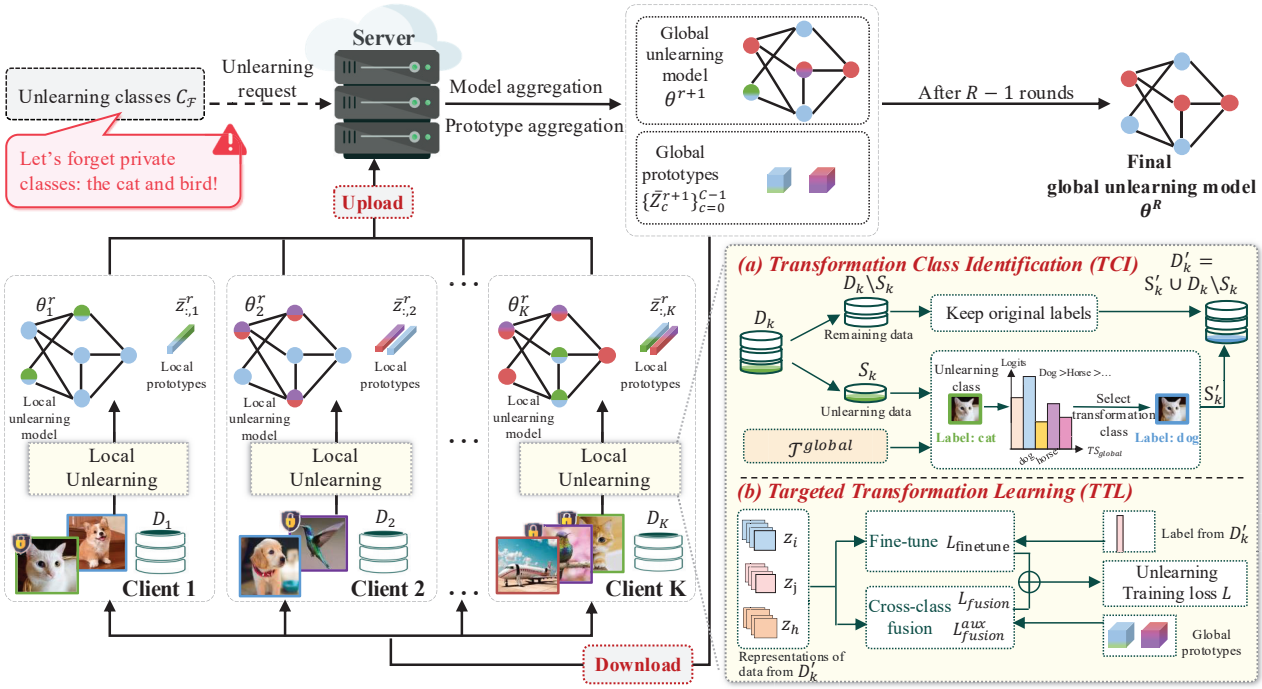

Figure 2. Illustration of our FUCRT framework. Clients perform local unlearning via TCI and TTL. The server aggregates local models and prototypes, then redistributes them to clients for the next round. After $R-1$ rounds, we obtain the final global unlearning model θ^R.

contains the samples to be forgotten.

3. Methodology

3.1. Overview

As illustrated in Fig. 2, our FUCRT maintains the standard FL architecture while introducing two components for FU: **(1) Transformation Class Identification (TCI).** We first generate the global transformation class set for unlearning classes, and then assign a target transformation label to each unlearning sample based on its highest output probability on this set. **(2) Targeted Transformation Learning (TTL).** We design a cross-class fusion mechanism to achieve integration between unlearning data and target transformation data within each client, which also ensures consistency of cross-class fusion across different clients under FL. The details are as follows.

3.2. Transformation Class Identification

Global Transformation Set Generation. Considering the original global model is typically well-trained in the FU setting, we identify candidate target transformation classes by examining the prediction probabilities for the unlearning class data, excluding the unlearning classes. In this way, we leverage the well-trained model's ability to recognize classes that share similar features with unlearning classes. Specifically, for each unlearning class $c_j \in \mathcal{C}_\mathcal{F}$, we first analyze the probability outputs of correctly classified samples of c_j on client k. We then sum these probabilities for each class and select those whose total probability exceeds a threshold to filters out dissimilar categories. Next, we obtain the selected target transformation classes to form a local transformation set $\mathcal{T}_{j,k}^{local}$ on client k for c_j.

After all clients compute their local transformation sets $\mathcal{T}_{j,k}^{local}(k \in \{1,\dots,K\})$ for each unlearning class c_j, they transmit these sets to the server, which aggregates them into global transformation sets $\mathcal{T}_j^{global}$ by maximum consensus. For details, the size of each global set $\mathcal{T}_j^{global}$ is determined by the most frequent set size among the local transformation sets. Then, the server selects the top $|\mathcal{T}_j^{global}|$ categories that appear most frequently across all local transformation sets to form $\mathcal{T}_j^{global}$. Based on this, the total global transformation set for all unlearning classes is represented by: $\mathcal{T}^{global} = \{\mathcal{T}_1^{global}, \mathcal{T}_2^{global}, \dots, \mathcal{T}_j^{global}, \dots\}$.

Target Transformation Label Identification. After obtaining $\mathcal{T}^{global}$, we identify target transformation labels from this set to unlearning samples. For each unlearning sample belonging to $\forall c_j \in \mathcal{C}_\mathcal{F}$, we identify the class with the highest output probability in $\mathcal{T}_j^{global}$ as the target transformation class. For an unlearning sample (x_i, y_i), this identification can denote as $(x_i, y_i) \rightarrow (x_i, y_i')$. Following this, we can get a transformation unlearning subset $\mathcal{S}_k' = (x_i, y_i')$ for client k. Meanwhile, we keep the labels of

Algorithm 1: FUCRT

Input: the original global model θ^0, unlearning classes $\mathcal{C}_\mathcal{F}$, local dataset $\mathcal{D}_k, k \in [K]$, K clients, and maximum round R.

Output: The final global unlearning model θ^R.

Procedure *Server execution*:
Generating the global transformation set $\mathcal{T}^{global}$;
for $r = 1$ *to* $R-1$ **do**
  for $k \leftarrow 1$ ***to*** K *in parallel* **do**
    send the global model θ^r to the client k;
    send the global class prototypes $\bar{Z}^r = (\bar{Z}^r_0, \bar{Z}^r_1, ..., \bar{Z}^r_{C-1})$to the client k;
    $\theta^r_k, \left\{\bar{z}^r_{c,k}\right\}_{c=0}^{C-1} \leftarrow$ *Local Unlearning* $(r,k,\theta^r,\bar{Z}^r)$
  end
  Obtain global unlearning model θ^{r+1} by aggregating r-th round local models by Eq.5;
  Obtain global prototype $\bar{Z}^{r+1}_c$ by aggregating local prototypes by Eq.6;
end
Return the final unlearning model θ^R.
Procedure *Local Unlearning*$(r, k, \theta^r, \bar{Z}^r)$:
Get the local model θ^r_k by θ^r: $\theta^r_k \leftarrow \theta^r$;
Obtain the new dataset $\mathcal{S}'_k \cup \mathcal{D}_k \backslash \mathcal{S}_k$ by using transformation class identification: $\mathcal{S}'_k \leftarrow \mathcal{S}_k$;
for *each batch* $\in \mathcal{S}'_k \cup \mathcal{D}_k \backslash \mathcal{S}_k$ **do**
  compute $\mathcal{L} = \mathcal{L}_{\text{finetune}} + \lambda_1 \mathcal{L}_{\text{fusion}} + \lambda_2 \mathcal{L}^{\text{aux}}_{\text{fusion}}$;
  $\theta^r_k \leftarrow \theta^r_k - \eta \nabla \mathcal{L}$;
end
generate local prototypes $\left\{\bar{z}^r_{c,k}\right\}_{c=0}^{C-1}$ by θ^r_k on $\mathcal{S}'_k \cup \mathcal{D}_k \backslash \mathcal{S}_k$;
Return θ^r_k and $\left\{\bar{z}^r_{c,k}\right\}_{c=0}^{C-1}$.

the remaining subset $\mathcal{D}_k \backslash \mathcal{S}_k$ unchanged. So we can obtain a new local dataset $\mathcal{D}'_k = \mathcal{S}'_k \cup \mathcal{D}_k \backslash \mathcal{S}_k$.

3.3. Targeted Transformation Learning

After identifying target transformation classes, we need to transform unlearning data from their original labels to their target classes, while preserving the model's discriminative performance on the remaining data. Note that all the following processes are based on $\mathcal{S}'_k \cup \mathcal{D}_k \backslash \mathcal{S}_k, k \in \{1, \cdots, K\}$.

Fine-tuning. For each client, we apply the cross-entropy loss to a batch of N samples:

$$\mathcal{L}_{\text{finetune}} = -\frac{1}{N}\Big(\sum_{i \in \mathcal{D}_k \backslash \mathcal{S}_k} y_i log(p) + \sum_{i \in \mathcal{S}'_k} y'_i log(p)\Big), \tag{1}$$

where p is the probability output predicted by the model for the sample $(x_i, y_i) \in \mathcal{D}_k \backslash \mathcal{S}_k$ and $(x_i, y'_i) \in \mathcal{S}'_k$.

Cross-Class Fusion. To further enhance model erasing guarantee, we propose a cross-class fusion mechanism, which integrates the unlearning data into the target transformation class data within each client. Inspired by traditional contrastive learning [23], we propose a cross-class contrastive learning, which not only pulls same-class instances closer while pushing different classes away, but also intentionally pulls certain different classes closer (i.e., unlearning classes toward their semantically similar remaining classes). This process can be formulated as follows:

$$\mathcal{L}_{\text{fusion}} = -\frac{1}{N}\sum_{i=1}^{N}\frac{1}{N_{y_i}}\sum_{j=1}^{N}\Big(\underbrace{\mathbf{1}_{y_i=y_j}\log\frac{e^{z_i \cdot z_j/\tau_t}}{\sum_{h=1}^{N}\mathbf{1}_{h\neq i}e^{z_i \cdot z_h/\tau_t}}}_{\text{Pull same class \& Push different classes}} + \underbrace{\mathbf{1}_{y_i=y'_j \& y'_i=y_j \& y'_i=y'_j}\log\frac{e^{z_i \cdot z_j/\tau_t}}{\sum_{h=1}^{N}\mathbf{1}_{h\neq i}e^{z_i \cdot z_h/\tau_t}}}_{\text{Cross-class fusion}}\Big), \tag{2}$$

where z_i, z_j, and z_h represent representations of input samples x_i, x_j, and x_h, respectively. y'_i (y'_j) is the target transformation label of unlearning sample x_i (x_j). For i-th sample, N_{y_i} represents the number of samples in the batch sharing the same label y_i. τ_t is the temperature scaling factor.

Moreover, considering the data heterogeneity across clients in FL, we introduce an auxiliary fusion loss to ensure consistency across different clients. Specifically, we compute local class prototypes within each client and aggregate them into global prototypes on the server. These global prototypes serve as bridges connecting different clients for cross client consistency, which can be formulated as:

$$\mathcal{L}^{\text{aux}}_{\text{fusion}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=0}^{C-1}\Big(\underbrace{\mathbf{1}_{y_i=j}\log\frac{e^{z_i \cdot \bar{Z}_j/\tau_t}}{\sum_{h=0}^{C-1}\mathbf{1}_{h\neq j}e^{z_i \cdot \bar{Z}_h/\tau_t}}}_{\text{Pull same class \& Push different classes}} + \underbrace{\mathbf{1}_{y'_i=j}\log\frac{e^{z_i \cdot \bar{Z}_j/\tau_t}}{\sum_{h=0}^{C-1}\mathbf{1}_{h\neq j}e^{z_i \cdot \bar{Z}_h/\tau_t}}}_{\text{Cross-class fusion}}\Big), \tag{3}$$

where $\bar{Z}_j$ is the global prototype of the j-th class.

3.4. Overall Unlearning Process

The overall unlearning process is shown in Algorithm 1. At the beginning of the first training round, the server receives the local transformation sets from clients and aggregates them into a global transformation set $\mathcal{T}^{global}$, as detailed in Section 3.2. Note that $\mathcal{T}^{global}$ is distributed back to all clients, and this process is performed only once.

On Local Client. In the r-th round, each client first identifies a target transformation class for each unlearning sample to construct a new local dataset $\mathcal{D}'_k$, where $k \in \{1, \ldots, K\}$. Then, all clients use $\mathcal{L}$ for the unlearning training to obtain their local unlearning model θ^r_k as:

$$\mathcal{L} = \mathcal{L}_{\text{finetune}} + \lambda_1 \mathcal{L}_{\text{fusion}} + \lambda_2 \mathcal{L}_{\text{fusion}}^{\text{aux}}, \tag{4}$$

where λ_1 and λ_2 are coefficients governing the contributions of $\mathcal{L}_{\text{fusion}}$ and $\mathcal{L}_{\text{fusion}}^{\text{aux}}$ losses, respectively.

On Federated Server. In the r-th round, the server aggregates the uploaded local unlearning models from clients into a global unlearning model as follows:

$$\theta^{r+1} = \frac{1}{K} \sum_{k=1}^{K} \theta_k^r, \tag{5}$$

where θ_k^r is the local unlearning model in client k. Meanwhile, the server obtains the global prototype $\bar{Z}_j^{r+1}$ by aggregating the local prototypes $\bar{z}_{j,k}^r$ from K clients:

$$\bar{Z}_j^{r+1} = \frac{1}{K} \sum_{k=1}^{K} \bar{z}_{j,k}^r, \tag{6}$$

where local prototype $\bar{z}_{j,k}^r$ is the average representation of class j on the client k in r-th round. After this round, the server sends the updated global unlearning model θ^{r+1} and global prototypes $\bar{z}_{j,k}^{r+1}$ to all clients for the next $r+1$ round.

4. Experimental Evaluation

4.1. Experimental Setup

Datasets and Network Architecture. We conduct extensive experiments on the following four real-world datasets: CIFAR10 [24], CIFAR100 [24], FMNIST [46], EuroSAT [21]. For the CIFAR10, FMNIST, and EuroSAT datasets, we employ the widely used ResNet18 as the default network architecture [20]. For CIFAR100, we opt for the more powerful ResNet50 as the network architecture.

Baselines. We adopt eight FU methods as our baselines: Retrain, Fine-tune, Gradient-ascent [45], FUDP [44], FUMD [51], VeriFi [10], FedAU [14], and FedOSD [36].

Federated Learning Setting. In our experiments, we set the total number of clients at 20, with all clients selected per round. The default Dirichlet coefficient δ=0.5 for the Non-IID scenario. We employ SGD [1] as the default optimizer.

Metrics. We evaluate erasing guarantees using two key metrics: (1) the accuracy of the unlearning model on the unlearning data, and (2) the attack success rate (ASR) obtained via membership inference attack (MIA) [39].

4.2. Main results

Erasing Guarantee. For an effective unlearning method, the unlearning model is expected to contain minimal information about the unlearning data. As shown in Table 1, we present a comparison of the accuracy and ASR of unlearning data by different federated unlearning methods on the CIFAR10, CIFAR100, FMNIST, and EuroSAT datasets. (1) Regarding the accuracy on unlearning data, the results illustrate that Fine-tune has the worst information erasure results, with its accuracy of U-set ranging from 0.11% to 52.04% on four datasets. It can be observed that the accuracy of our method on U-set is the same as the Retrain, and both achieve effective erasure, i.e., 0% accuracy on unlearning data. (2) From the perspective of ASR from MIA, the results present that FedAU has the weakest information erasure results, with its ASR ranging from 69.14% to 99.80% on four datasets. This is because FedAU primarily achieves unlearning by adjusting the fully connected layers of the model, causing minimal effect in the representation space. We can find that the ASR of our method is also the same as the Retrain, and both achieve effective erasure, i.e., 0% ASR. These results prove that our method achieves the same level of information erasure from the pre-trained original model as the Retrain does.

Model Utility Preservation. As presented in Table 1, we also evaluate the model utility preservation through the accuracy of remaining data. The results indicate that our proposed FUCRT achieves higher accuracy on remaining data than all baselines (Gradient-ascent, FUDP, FUMD, VeriFi, FedAU, and FedOSD) that effectively achieved forgetting. On the IID settings, our method outperforms Gradient-ascent, FUDP, FUMD, VeriFi, FedAU, and FedOSD by 78.70%, 3.54%, 20.48%, 9.95%, 4.79%, and 2.96% respectively in average accuracy of the remaining data across four datasets. Similarly, on the Non-IID settings, our method surpasses Gradient-ascent, FUDP, FUMD, VeriFi, FedAU, and FedOSD by 77.51%, 5.37%, 19.90%, 12.61%, 20.06%, 3.87% respectively. Fine-tune improves accuracy on remaining data compared to Retrain in most cases. However, it fails significantly at erasing information from the unlearning data. The accuracy of our method on the remaining data is closest to that of the Retrain method, with a minimal average difference of only 0.11% on the four datasets. In some cases, our method performs even better than Retrain. This is because these baselines focus on identifying and modifying original model parameters relevant to unlearning data. The complex parameter interactions in deep neural networks make it difficult to identify and isolate parameters related to unlearning data from those associated with remaining data. As a result, when performing effective erasure, these methods will inevitably degrade the performance of the remaining data, i.e., impacting the utility preservation. Different from previous approaches, the core of FUCRT is the transformation. Our method allows the model to adaptively adjust its parameters, achieving 100% erasure while better preserving utility on the remaining data.

Reducing Unlearning Time. In Table 1, we further evaluate the average time cost required for each unlearning method to complete the forgetting process across four datasets. The results indicate that Retrain takes the longest average time in both IID and Non-IID settings. We can ob-

Table 1. Accuracy(%) and ASR (%) on CIFAR10, CIFAR100, FMNIST, and EuroSAT datasets when unlearning 10% classes, with average time cost (s) across four datasets. "U-set" and "R-set" denote the unlearning data and remaining data in the test set, respectively.

	Method	CIFAR10			CIFAR100			FMNIST			EuroSAT			Ave.
		U-set	R-set	ASR	U-set	R-set	ASR	U-set	R-set	ASR	U-set	R-set	ASR	Time(s)
IID	Original Model	93.56	87.70	94.66	75.48	75.88	72.69	98.46	89.21	98.51	94.09	94.70	99.55	—
	Retrain	0.00	89.37	0.00	0.00	76.73	0.00	0.00	91.35	0.00	0.00	94.94	0.00	1925
	Fine-tune	52.04	89.82	58.78	49.33	76.38	45.06	91.79	91.34	95.70	1.55	96.33	7.40	1269
	Gradient-ascent [45]	0.00	10.94	0.00	0.00	5.08	0.00	0.00	7.92	0.00	0.00	12.31	0.00	85
	FUDP [44]	0.00	84.85	0.00	0.03	72.25	0.00	0.00	89.31	0.00	0.00	90.47	0.00	1135
	FUMD [51]	0.00	82.38	0.00	0.00	8.45	0.00	0.00	88.41	0.00	0.00	89.90	0.00	96
	VeriFi [10]	0.00	83.57	0.00	0.00	75.02	0.00	0.00	75.90	0.00	0.00	76.75	0.00	101
	FedAU [14]	0.00	88.75	93.79	0.00	72.41	68.85	0.00	83.07	98.42	0.00	87.68	97.71	112
	FedOSD [36]	0.00	88.30	0.00	0.00	74.54	3.43	0.00	90.57	3.09	0.00	85.78	0.96	89
	FUCRT (Ours)	0.00	90.06	0.00	0.00	75.25	0.00	0.00	90.82	0.00	0.00	94.92	0.00	68
Non-IID	Original Model	91.77	86.14	96.15	75.54	75.65	72.44	98.87	86.59	98.95	87.55	84.40	98.76	—
	Retrain	0.00	88.95	0.00	0.00	77.00	0.00	0.00	89.94	0.00	0.00	86.69	0.00	2238
	Fine-tune	17.16	90.08	92.90	48.98	76.59	48.58	30.47	91.19	97.81	0.11	92.58	9.33	1515
	Gradient-ascent [45]	0.00	11.15	0.00	0.00	3.01	0.00	0.00	8.80	0.69	0.00	10.02	0.00	101
	FUDP [44]	0.00	85.68	0.00	8.88	69.71	0.00	0.00	85.17	0.00	0.00	81.00	0.00	1415
	FUMD [51]	0.00	82.60	4.02	0.00	8.95	0.00	0.00	88.86	0.10	0.00	83.01	0.00	116
	VeriFi [10]	0.00	85.79	18.19	0.00	72.92	0.00	0.00	67.64	0.00	0.00	66.24	2.80	121
	FedAU [14]	0.00	69.59	95.90	0.00	72.49	70.77	0.00	57.08	98.81	0.00	63.65	97.22	142
	FedOSD [36]	0.00	87.61	34.36	0.00	73.15	0.91	0.00	88.07	27.03	0.00	78.74	24.40	103
	FUCRT (Ours)	0.00	89.82	0.00	0.00	74.88	0.00	0.00	90.95	0.00	0.00	87.38	0.00	82

serve that FUCRT achieves the minimal average time cost for forgetting, taking 68s for IID settings and 82s for Non-IID settings. Our proposed FUCRT reduces unlearning time by $19.13\% \sim 96.38\%$ compared to other methods. This is because the representation transformation provides a direct unlearning path for unlearning data in the representation space such that performing representation transformation consumes little time once the target transformation classes of unlearning data are identified.

4.3. Ablation Studies

To further analyze FUCRT, we conduct ablation studies and results are illustrated in Table 2, where the accuracy on unlearning data is omitted due to value is 0 in different cases.

Effectiveness of TCI. We replace TCI with two random identification strategies: $\texttt{Random}_{\texttt{class}}$ which assigns a random class to all unlearning samples within each class, and $\texttt{Random}_{\texttt{sample}}$ which assigns a random class to each individual unlearning sample. We observe decreases in the accuracy of remaining data, e.g., $\texttt{Random}_{\texttt{class}}$ and $\texttt{Random}_{\texttt{sample}}$ have average accuracy declines of 3.11% and 0.69% across four datasets, respectively. Beyond accuracy degradation, these random strategies also compromise erasing guarantee, with MIA yielding higher success rates, e.g., ASR increases of 7.30% for $\texttt{Random}_{\texttt{class}}$ and 9.61% for $\texttt{Random}_{\texttt{sample}}$ across four datasets. These dual negative impacts demonstrate that random identification strategies are inadequate for both utility preservation and erasing guarantee. In contrast, our TCI outperforms random strategies by identifying more appropriate transformation classes through the exploitation of both class similarities and feature associations.

Effectiveness of TTL. TTL devises a cross-class fusion mechanism to achieve integration between unlearning data and target transformation class data, which also incorporates auxiliary cross-client fusion to ensure consistent unlearning across clients. We examine cross-class fusion by FUCRT w/o `fusion`, which means removing the cross-class fusion loss. To further illustrate the necessity of auxiliary cross-client fusion in this process, we also conduct FUCRT w/o `aux-fusion`, which means removing the auxiliary cross-client fusion loss. w/o `fusion` reduces the accuracy on remaining data by 0.30%$\sim$2.78% and increases the ASR from MIA by 0.00%$\sim$65.70% on four datasets. This means that after transformation class identification, only fine-tuning can achieve competitive accuracy on both remaining and unlearning data relative to the baselines. w/o `aux-fusion` reduces the accuracy on remaining data by 0.07%$\sim$4.42% and increases the ASR from MIA by 0.00%$\sim$3.86% on four datasets. This indicates that the auxiliary fusion loss can ensure consistency across different clients by increasing the accuracy on remaining data while reducing ASR, particularly in Non-IID settings. These results highlight the importance of cross-class fusion beyond fine-tuning, effectively integrating unlearning data and target transformation classes to ensure erasure guarantees while improving accuracy on remaining data.

4.4. Analysis

Impact of Data Heterogeneity. To study the impact of data heterogeneity on the performance of our method, we

Table 2. Ablation studies on the accuracy of remaining data and ASR on four datasets with unlearning 10% classes.

Method	R-set Accuracy								MIA ASR							
	CIFAR10		CIFAR100		FMNIST		EuroSAT		CIFAR10		CIFAR100		FMNIST		EuroSAT	
	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID
FUCRT(Ours)	**90.06**	**89.82**	**75.25**	**74.88**	**90.82**	**90.95**	**94.92**	**87.38**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
TCI → $\text{Random}_{\text{class}}$	88.36	88.24	73.13	73.61	87.78	85.19	89.17	83.72	2.61	9.10	0.00	5.35	1.50	15.60	6.34	17.88
Δ	1.70 ↓	1.58 ↓	2.12 ↓	1.27 ↓	3.04 ↓	5.76 ↓	5.75 ↓	3.66 ↓	2.61 ↑	9.10 ↑	0.00 ↑	5.35 ↑	1.50 ↑	15.60 ↑	6.34 ↑	17.88 ↑
TCI → $\text{Random}_{\text{sample}}$	89.73	89.14	74.81	74.61	90.34	90.26	94.20	85.45	9.82	12.33	0.68	9.81	7.82	17.47	6.20	12.74
Δ	0.33 ↓	0.68 ↓	0.44 ↓	0.27 ↓	0.48 ↓	0.69 ↓	0.72 ↓	1.93 ↓	9.82 ↑	12.33 ↑	0.68 ↑	9.81 ↑	7.82 ↑	17.47 ↑	6.20 ↑	12.74 ↑
w/o `fusion`	89.28	88.38	74.95	73.13	89.52	89.41	92.14	85.87	4.87	65.70	0.00	32.10	0.31	10.83	7.88	30.93
Δ	0.78 ↓	1.44 ↓	0.30 ↓	1.75 ↓	1.30 ↓	1.54 ↓	2.78 ↓	1.51 ↓	4.87 ↑	65.70 ↑	0.00 ↑	32.10 ↑	0.31 ↑	10.83 ↑	7.88 ↑	30.93 ↑
w/o `aux-fusion`	89.90	89.59	75.01	74.75	90.75	89.93	90.50	84.41	0.00	1.71	0.30	2.30	0.00	3.86	0.00	1.56
Δ	0.16 ↓	0.23 ↓	0.24 ↓	0.13 ↓	0.07 ↓	1.02 ↓	4.42 ↓	2.97 ↓	0.00 ↑	1.71 ↑	0.30 ↑	2.30 ↑	0.00 ↑	3.86 ↑	0.00 ↑	1.56 ↑

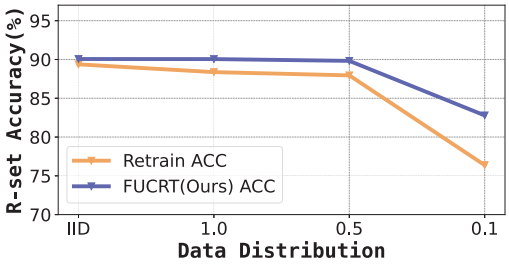

Figure 3. The impact of different data distributions (δ) on accuracy of the remaining data under CIFAR10.

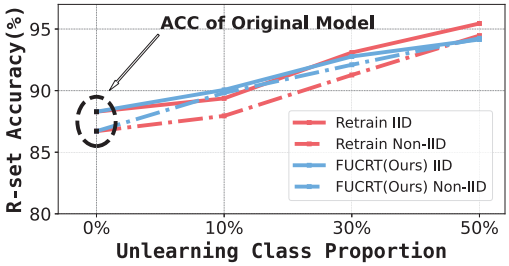

Figure 4. The impact of class proportions on accuracy of the remaining data under CIFAR10. Here, 10%, 30%, and 50% represent unlearning data with 1, 3, and 5 classes, respectively.

compared FUCRT and Retrain under different data distributions, as illustrated in Figure 3. We plotted the accuracy of the remaining data for four heterogeneity levels. We omit the results on unlearning data, where both FUCRT and Retrain achieve 0% accuracy. In Figure 3, as heterogeneity increases, differences in local representation space distributions among clients grow, which negatively impacts the accuracy of the remaining data. Our method handles heterogeneity scenarios better than Retrain. This is because FUCRT's cross-class fusion mechanism aligns local representation spaces across clients, reducing conflicts between class-aware representations.

Impact of Unlearning Class Proportion. We conducted experiments on the CIFAR10 dataset with unlearning proportions of 10%, 30%, and 50% classes. As presented in Figure 4, we compare FUCRT to the Retrain method on the accuracy of remaining data. We observe that for both FUCRT and Retrain methods, the accuracy increased as the proportion of unlearning classes grew. The reason behind this trend is that fewer remaining classes result in a representation space that is easier to construct and optimize. Overall, FUCRT performs comparably to or better than Retrain across different unlearning proportions.

4.5. Visualization of Representation Space

To better illustrate the differences in the representation space across various unlearning methods, we employed t-SNE [42] to visualize the representation space on CIFAR10 with Non-IID settings, as illustrated in Figure 5. The results indicate that our proposed FUCRT exhibits a pattern most similar to the Retrain method. In the representation space of FUCRT and Retrain, the representation of "automobile" is transferred to the representation of "truck", and the representations of both become a cluster. Similarly, "cat" is transferred to "dog". Other baselines do not show the same pattern as Retrain method due to the limitations of their respective methods. Moreover, the clusters of FUCRT appear more compact compared to Retrain. This can be attributed to the cross-class fusion, which optimizes the distribution of the representation space, amplifying inter-class differences while minimizing intra-class variance. This observation aligns with our previous experimental results, where FUCRT slightly outperforms the Retrain method on the accuracy of the remaining data.

5. Related Works

Federated Learning. The foundational work in FL is the FedAvg [33], which aggregates locally computed model updates to form a global model. FL has been widely used in privacy-sensitive fields such as healthcare [30, 35], finance [19, 38], and recommendation [49]. However, FL faces numerous challenges [15, 17, 22], among which the most fundamental is data heterogeneity [16, 26, 32, 34]. Data heterogeneity means data distributions among clients

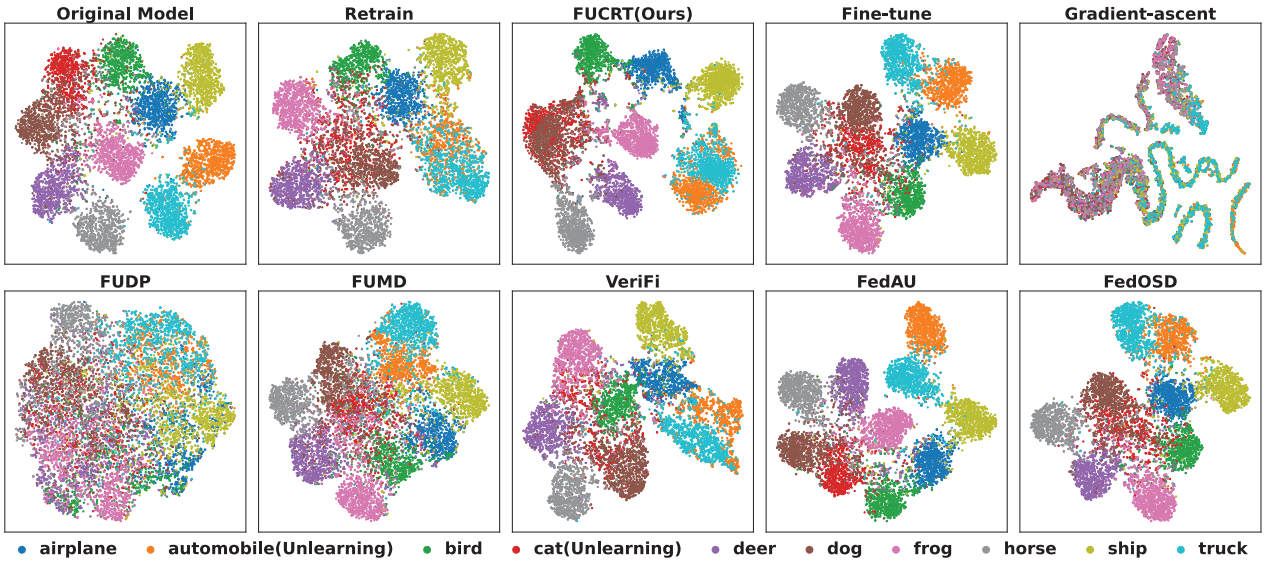

Figure 5. t-SNE visualization of the representation space of different unlearning methods on CIFAR10 with the Non-IID setting.

are often not independent and identically distributed (Non-IID), which is also a key issue faced in federated unlearning.

Machine Unlearning. The concept of machine unlearning was first introduced by Cao *et al.* [6], with the aim of developing techniques to more efficiently and accurately erase specific information from the well-trained model, thereby protecting the privacy of data holders [18, 25, 41]. The most straightforward approach is to remove sensitive data and retrain the model from scratch. SISA [4] is a pioneering approach that trades extra storage space for faster unlearning time. Another unlearning method involves altering the model using precomputed parameters to adjust the target model [3, 28]. Additionally, unlearning can also be achieved by pruning certain gradients from a trained model [5, 8, 37]. However, current centralized unlearning methods struggle with FL challenges, such as limited data access and Non-IID data.

Federated Unlearning. FU frameworks can be categorized into three distinct levels based on the specific targets [27, 31, 40, 45]: client-level, sample-level, and class-level. Client-level FU seeks to remove the knowledge contributed by specific clients from the global model [10, 28, 31, 36]. Sample-level FU aims to eliminate knowledge derived from particular subsets of client data [7, 47]. Class-level FU aims to remove specific classes from the generalization boundary of FL model [44, 51]. Recent studies usually identify and modify original model parameters relevant to unlearning data for FU. For example, FUDP [44] prunes the most relevant channels associated with the classes to be forgotten. FFMU [7] uses nonlinear functional theory to improve the efficiency while maintaining the unlearning quality. FedRecovery [50] is an efficient unlearning algorithm that leverages differential privacy to erase client impact by removing gradient residuals from the model. FUMD [51] utilizes a randomly initialized degradation model to erase the existence of the unlearning classes. Similarly, FedAU [14] employs an auxiliary unlearning classifier combined with linear operations to facilitate unlearning. FedOSD [36] establishes an orthogonal steepest descent direction to accelerate the unlearning process while mitigating the model utility reduction. Unlike these approaches, our FUCRT allows the model to adaptively adjust its parameters, achieving efficient erasure while better preserving model utility on the remaining data.

6. Conclusion

In this work, we propose a novel idea for FU, that is, transforming the distribution of unlearning data to fuse with their semantically related remaining data in the representation space. Based on this idea, we develop a new framework, FUCRT, to achieve effective FU. FUCRT contains two main components: First, it introduces a transformation class identification strategy, that leverages the original model to identify appropriate transformation classes for unlearning data. Second, it employs a targeted transformation learning process to ensure the effective and consistent transformation of unlearning data. Extensive experiments on four datasets demonstrate that FUCRT not only achieves effective data erasure but also outperforms state-of-the-art methods by an average of 2.96%/3.78% in utility preservation under IID/Non-IID settings, while reducing unlearning time by 19.13%∼96.38%.

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